Grocery Shopping Optimization Using Dynamic Programming (0/1 Knapsack)

Team Composition

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Project Description

When shopping for groceries, I often find myself standing in the store unsure of what to buy. Most of the time I have certain nutritional goals in mind such as getting enough protein, vitamins, or minerals like magnesium, but comparing multiple products and keeping track of their nutritional values can quickly become overwhelming. Because of this, I sometimes end up purchasing items that do not fully meet my nutritional needs, or I overcompensate by buying too much of one type of nutrient while lacking others.

This is a common challenge for many people who are trying to eat healthier while also staying within a budget. There is usually a tradeoff between cost and nutrition, and determining the best combination of grocery items is not always straightforward when evaluating dozens of products in a store.

This project aims to address this problem by developing an algorithm that helps users select the best combination of grocery items based on both their budget and nutritional goals. The problem can be modeled using the 0/1 Knapsack dynamic programming algorithm, where each grocery item has an associated cost and a nutritional value score. The goal is to maximize the overall nutritional value of the selected items while ensuring that the total cost does not exceed a fixed budget.

By applying dynamic programming, the algorithm evaluates different combinations of items and identifies the optimal selection that provides the highest nutritional benefit within the budget constraint.

The final system could potentially be implemented as a mobile or in store application that assists users while shopping. Such a system could guide users toward healthier purchasing decisions while ensuring they stay within their financial limits. This demonstrates how algorithmic techniques such as dynamic programming can be applied to everyday decision-making problems.

Dataset Description

For this project, we will use the Open Food Facts dataset, which is a large open source and community driven database of food products collected from around the world. The dataset contains information on over four million food products across more than 150 countries, making it highly representative of real-world grocery store items.

Each product entry includes detailed attributes such as product name, brand, ingredients, allergens, and most importantly comprehensive nutritional information. These nutritional attributes include macronutrients such as protein, fats, and carbohydrates, as well as micronutrients such as vitamins and minerals including magnesium, calcium, and iron.

The dataset is structured as a large table containing more than 150 columns, covering multiple aspects such as product identification, nutritional composition per 100g, ingredient lists, and classifications such as Nutri Score. This level of detail allows us to model realistic decision making scenarios where shoppers must balance multiple nutritional factors when selecting food products.

However, due to the size and complexity of the dataset, significant preprocessing will be required before applying any algorithms. First, the dataset will be reduced to a manageable subset by selecting only the columns relevant to this project, such as product name, nutritional values such as protein, calories, vitamins, and key minerals, and price if available or an assigned proxy value.

Next, missing data will be analyzed and handled appropriately. Missing values may occur under different conditions such as Missing Completely at Random, Missing at Random, or Missing Not at Random. Depending on the pattern, strategies such as removal, imputation, or substitution will be applied. Rows containing excessive missing values or inconsistent entries will be filtered out.

Additional preprocessing steps will include removing duplicate records, handling null values, standardizing units for example ensuring nutritional values are consistently measured per 100g, and normalizing data so that nutritional values can be compared across products.

After preprocessing, the cleaned dataset will provide a structured and reliable input for applying the dynamic programming algorithm to optimize grocery selection based on nutritional value and budget constraints.

Kaggle dataset link - https://www.kaggle.com/datasets/openfoodfacts/world-food-facts

Official dataset link - https://world.openfoodfacts.org/data

Implementation

The implementation will be developed in Python and will focus on solving the grocery selection problem using the 0/1 Knapsack dynamic programming algorithm.

The workflow for the project will include the following steps:

1. Data preprocessing - The Open Food Facts dataset will be cleaned and reduced to relevant nutritional and product attributes. Missing values and inconsistencies will be handled during this step.
2. Nutritional value scoring - Each item will be assigned a value score derived from selected nutritional attributes. This score will represent the benefit associated with including that item in the grocery basket.
3. Dynamic programming optimization - The 0/1 knapsack algorithm will be implemented to determine the optimal set of grocery items that maximizes total nutritional value while staying within a specified budget.
4. User input and optimization - Users will be able to specify a budget such as $30 or $50. The algorithm will then compute the optimal selection of grocery items within that constraint.
5. Output presentation - The program will output the selected items along with the total cost and total nutritional value score, allowing users to understand how the algorithm optimized their grocery choices.

Responsibilities

This project will be completed by a two member team with responsibilities divided across data processing, algorithm implementation, and system development.

- Simran Kharbanda

Responsible for dataset preprocessing, cleaning the Open Food Facts dataset, handling missing values, selecting relevant nutritional attributes, and preparing the dataset for analysis. Also responsible for documentation and managing the final project repository.

- Shashank Ashoka

Responsible for implementing the 0/1 knapsack dynamic programming algorithm, developing the optimization logic, integrating user inputs such as budget constraints, and generating the final optimized grocery selection results.